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Augmented Reality and Internet of Things Trends in Education

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***Abstract*—Augmented Reality (AR) Technologies are gaining considerable attention due to their increased stability and reliability. Combined with the availability of smarter and cheaper devices, like smart-phones and tablets, AR technology greatly helps in enhancing the amount, relevance and ease of access of the available information. At the same pace, Internet of Things (IoT) technologies due to their pervasiveness, are paving the way towards a smarter interconnected world. Several industries, like entertainment, medicine and engineering, have developed and adapted IoT and AR technologies where the enrichment of the physical reality by the information and capabilities offered by AR proved useful for their business. The same approach should and could be the case in the education field. Nevertheless, the usage of IoT and AR technologies is still in its infancy, not following yet the path of many other industry fields. In this paper, after the presentation of contemporary AR and IoT technologies a proposal for utilizing these technologies to teach concepts that are required by the curriculum of the "Informatics Applications" course for the first class of senior high schools in Greece is presented.**

Keywords—Education, Augmented Reality, Internet of Things

I. Introduction

The rapid growth of technology, including the Internet, smartphones, and mobile applications, has changed drastically the users' daily interactions. One of these emerging technologies is Augmented Reality (AR) [1], which can offer a new way in exchanging information and ideas in an interesting and attractive manner [2].

The wide spread of smartphone devices with advanced features has introduced new technologies, like the Mobile Augmented Reality (Mobile AR). The use of AR technology provides an enhanced depiction of reality combining the real-world with virtual three-dimensional (3D) elements [3]. More specifically, AR apps have already been embedded into smartphone devices with the purpose to provide interactive experiences to the users [4].

Furthermore, a transformation of the business environments is also observed given the wide spread of smart technologies in the peoples' everyday lives. There is a huge change on how people interact in their daily purchases and exchanges. Smart retail and mobile apps offer customers new, more complete, exciting and interesting experiences by integrating AR technologies in their user daily interactions [5].

Despite the above mentioned scenery changes, when it comes to education, AR has been spreading and gaining attention, mostly towards video games and entertainment, but has yet a long way to penetrate fully into the education field where it can play an important role [2].

Similarly, the IoT technologies are present in almost everyone's life via the widespread use of smart devices and the various services offered by the numerous available applications. The IoT systems have entered and they have shaped already existing fields into new more efficient counterparts. The prefix word "smart" comes to identify such fields. To mention a few, smart homes, smart buildings, smart cities, smart metering, smart healthcare, smart industrial control, smart retail, smart supply chains and smart logistics are some such fields [6]. The early adoption of IoT in so many fields has highlighted the importance of the research towards the adoption of IoT systems in education as well. A few small steps have been made so far towards this direction with networked computers, smartphones and tablets not to be yet embedded in the everyday educational practice. Interactive whiteboards are entering slowly into the classrooms, mostly due to their cost and the lack of the necessary networking infrastructure, but even when they are present, they are not yet fully utilized. Thankfully, robotic kits and IoT project-based activities are being introduced in primary, secondary and tertiary education slowly but steadily [7].

Given the above, researchers come to the aid proposing adoption ideas in the education sector. There are many research papers recommending ideas, regarding the adoption of IoT systems, that improve the efficiency of the education "business" and environment as well as proposals about improving the way teaching and learning takes place in the educational practice.

Regarding the first part, many IoT researchers have invested significant efforts proposing ways to improve the energy efficiency of educational buildings and classrooms [8][9], approaches to deal with campus/school safety and surveillance [10][11][12], students' auto-attendance and behavior reporting [13][14], and even ways to address the automatic guidance towards teaching venues in large campuses [15]. All these proposals indirectly ensure a more efficient educational environment, reduce cost, automate day-to-day repeated tasks performed by teachers and personnel, help in monitoring buildings and students, and

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leave more time for the main concern of the educational institutes, which is efficient teaching and learning.

There are also several proposals regarding the improvement of learning and teaching utilizing IoT technology. Indicatively, the IoT in Educational context research literature deals with matters like helping the process of language teaching and learning [16], a way of improving classroom activities by [17], the development of environmental awareness [18][19], and the assistance of physical education and training [20].

The utilization of IoT technology is especially assistive in teaching and learning information and communication technologies (ICT) subjects as it comprises a wide area of subjects such as programming, microcontrollers and generally electronics related subjects, databases, big data, machine learning, cloud technology, and networking [21]. Researchers in this area recommend ways of teaching concepts and skills using the IoT technology as a way to teach and assimilate knowledge by applying and practicing in a more context specific way [22]. Since teaching and learning is the main subject matter in education, modern ideas of instructional design have been proposed as well [23][24][25].

There are also several research works recommending the combined utilization of AR technology and IoT systems in education mainly for enriching the provided information and for widening the way of interacting with the educational context [26]. This combination achieves a more engaging, safe and effective path towards learning [27][28]. Similarly, a more efficient way of teaching, by applying knowledge and developing skills, is offered via providing an augmented way (i.e. through AR) of interacting with the physical world interfaces (i.e. through IoT) [29].

In this work, a series of project-based activities using the Science Technology Engineering Mathematics (STEM) approach is proposed for the teaching of the course "Informatics Application" of the first class of the senior high school. The proposed activities have the goal of getting the students accustomed with the concepts of IoT and AR technologies while following the guidelines described in the course's curriculum. Concepts like 3D design, programming, mobile app creation, robotics and even machine learning can be addressed with the proposed series of activities, thus, leading in a more intriguing way of learning.

In the next paragraphs the following are included: in section II, we present the AR technologies including their relation with education. In section III, we include an IoT technologies' approach and how it can be implemented in educational settings. In section IV, a combination of the two technologies, i.e. of AR and IoT, is presented and in section V, a proposal of AR an IoT technologies for educational use is suggested. Lastly, conclusions and future steps of this research are included in section VI.

II. Augmented Reality (AR)

A. Augmented Reality (AR) Technologies

Augmented Reality (AR) can be defined as an atmosphere of blended reality in which virtual data is added on the real-world image simultaneously [30] or as an immersive environment through which digital content is displayed over the real-world, using a technological medium (such as a smartphone or a tablet computer). AR and VR (Virtual Reality) share a lot of similarities but their basic difference is that in AR the real world is mixed with virtual elements and the creation of a realistic virtual world as a real world illusion is not required [31]. In AR technology, the additional virtual content is somehow delimited and has specific boundaries in the real world in order to provide information and new experiences that extend the users' senses [32].

The mobile devices' sensors are a very important aspect in order for a mobile AR application to fully capture the surrounding environment and to place interactive objects to it. That offers a different and digitally enhanced representation of reality that can attract the user's interest [33]. Because of this kind of information representation, the usage of AR technology has increased over the last years especially for advertising purposes [34].

Of course, along with the expansion of the AR relative technology, an expansion of platforms for creating applications with AR technology can be seen. There are many available options such as ARToolKit [47], ARCore [48], Unity3D [49]. Their basic actions involve the marking of the position of a device using the sensor technology and the ability to create the 3D representation into the real world. This is a difficult task, which includes actions of "understanding" different shapes in the real world and their dimensions in relation to other factors, such as movement, orientation, lighting, and shadows, in order to create the effect of reality [35].

How the user experiences the final outcome of an application defines the degree of adoption of AR technology [36] and, of course, the available internet connections and device technologies play a significant part as well.

B. Augmented Reality (AR) Technologies in Education

The research that has been done over the last years towards the Augmented Reality (AR) direction provides valuable insights about its use and implementation in education.

In the study by Alfaro and Puyvelde [36] on a total of 1,752 user reviews about a set of 116 educational mobile AR apps available on the Google play store, the results suggest that the developers of educational mobile AR apps need to solve technical problems, improve certain features, and provide more explicit instructions to the users. Regardless of these needs, the users recognize that these apps have a great potential as educational tools. Tanalol et al. [37] conducted a study on user experience of AR applications in education. Sixteen students between the ages of 6 to 9 years old participated filling a response form in order to gather data about the usability and interactivity of AR apps. The results indicated that learning with the use of AR technology is more interesting and efficient in capturing the students' attention and in promoting STEM learning. Suhaimi et al. [38] created the Jawi-AR App, which is a smartphone application that integrates AR and was designed for primary school students. A total of 157 second year students from a primary school participated in their study. The results indicated that the app can help to improve the students' understanding and skills while changing the way students learn by promoting self-learning techniques.

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Fidan and Tuncel [39] investigated in their study the effects of Problem Based Learning (PBL) assisted with Augmented Reality (AR) on learning achievement and the students' attitude towards physics subjects as a part of science education. The FenAR software was developed to support the activities in the classroom with the participation of 91 students of the seventh grade. The results indicated that integrating AR into PBL activities both increased the students' learning achievement and promoted their positive attitudes towards physics subjects because they were more useful, realistic, and interesting and helped them to understand and analyze the problems that had to be solved in the educational scenarios. In the study of Auliya and Munasiah [40], the impact of using AR in learning 3D geometry on the students' attitudes and conceptual understanding was investigated. The research was carried out in a public secondary school in Indonesia with the participation of 61 students in total. The results indicated that the students' conceptual understanding in mathematics and attitude towards geometry in the AR group was better.

Given the rapid technological evolution, the characteristics of stability and reliability of AR technologies are obvious. This fact results in their rapid expansion and use, mainly in commercial settings. On the other hand and despite the several attempts of implementing AR in education, the adoption of educational applications of AR technology in education is still at a low level [34] and with limited use although AR is implemented with great success in many aspects of everyday lives [41]. One reason is the limited knowledge of developing AR applications by teachers and learners [32] or even the knowledge of how to use and implement them into their lessons despite the memorable efforts that already have been made resulting to the improvement in learning outcomes [42].

III. INTERNET OF THINGS (IOT)

A. *Internet of Things (IoT) Technologies*

Internet of Things (IoT) is defined by Oracle [43] as the network of physical objects with embedded sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet. Internet Engineering Task Force (IETF) [44] defines IoT as the network of physical objects or "things" embedded with electronics, software, sensors, actuators, and connectivity to enable objects to exchange data with the manufacturer, operator, and/or other connected devices. As it is obvious, these physical objects should have some kind of software running on them to be able to process and exchange mostly sensor data over the desired communication channels, which might be the Internet or another communication network.

These "things", alternatively stated as physical objects with processing ability, could be devices such as ordinary household objects to sophisticated industrial tools which are able to interact with the physical world through sensors and actuators. As stated in [43], the IoT end devices ("things" again) number is expected to reach 22 billion by 2025. Usually these "things" are resource-constraint devices since a large number of low-cost objects are usually required to be deployed, doing unattended simple things in distant places while being easily maintained or no maintained at all.

As it can be concluded from the above, the main constituent parts of the IoT systems are the "things", the interconnecting network and the exchanged data. Data are generated by or are needed for these interactions. These might be telemetry data (e.g. temperature, humidity, position) or control data (e.g. device or network configuration data and commands). Hence, the data should be sent or received and, thus, a communication path is needed. Usually, wireless IoT protocols are employed to interconnect the "things" to other "things" and/or to the necessary "central" infrastructure of the IoT systems. In combination with these, mostly IoT specific and not so resource demanding wireless protocols, the "lighter" counterparts of the standard network/Internet protocols over wired, wireless, or cell infrastructures are also used. Intermediate devices (Gateways) could be present in order to collect and forward data and, hence, to bridge these two worlds. Furthermore, network control logic could be partially or fully realized in the "things". In the first case, the network coordination could be happening in cooperation with another machine or instance/s in a cloud system (i.e. network controllers).

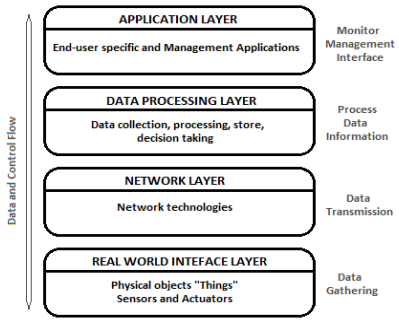

Fig. 1. IoT architecture.

Two more constituents/layers are present in the architecture of an IoT system, as depicted in Figure 1, i.e. the data processing layer and the application layer. The former is responsible for data collection, preprocessing, storing and for possible desired automated or even machine learning based decisions. The latter is the layer where the users'/customers' applications and the management applications of the IoT systems reside.

B. *Internet of Things (IoT) Technologies in Education*

There are several works regarding the usage of IoT in education showing that the stakeholders of education have already started taking into account IoT technology.

Diddeniya et al. [8] created an energy consumption control IoT system to manage and achieve ideal lighting and temperature conditions in the classroom considering also the human presence by means of the Microsoft Kinect technology. A similar concept has been dealt with in [9] by Mataloto et al. Their proposal is based on the LoRaWAN technology and certain heuristics are used to automate the energy/cost saving procedure.

Ahmed et al. in [10] proposed an IoT system that ensures the students' embarkation in a school bus with the use of RFIDs and fingerprint scanning and also gives parents the ability to track the school bus (via GPS technology) and receive notifications when the school bus is getting near home via an android smartphone application. An android

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application is also employed in [11] by Kumari et al. with the ability of real time monitoring of a child's position via RFIDs and a school bus's location via GSM and GPS technology along with other various data, such as available seats and possible incidents. In [12] the utilization of machine learning algorithms to process the video data to export identification and behavior inferences is tested. As Elbasi [12] mentions, abnormal behavior can be pinpointed in various environments by employing machine learning algorithms to process video data to export identification and behavior inferences.

In [14] a framework for the use of an IoT system is presented in parallel with a Learning Management System (LMS). Single board computers, cameras and machine learning algorithms are utilized to measure the students' activities, behavior and attentiveness in the classroom by observing their facial expression, thus assessing learning and teaching results. Yang et al. [13] created and tested a classroom attendance tracking IoT system. Using a smartphone-based application the students can declare their presence in a lecture using a voiceprint verification system (Azure cognitive services) and real-time location detection (GPS, cellular and WiFi signals). A corresponding phone application and a web environment were also created for the attendance management and the monitoring parts. Torres-Sospedra et al. [15] developed and tested a system able to help users find their way around a campus via smart phone applications. A novel indoor positioning system (IPS) is used that, combined with an outdoor position system, can support campus wayfinding. The application interface is an intuitive map-based view.

Regarding IoT in the educational process, many researchers have integrated IoT applications in their instructional design and tested their affordances in teaching and learning. Elena de la Guia et al. [16] created and tested an IoT system for foreign language teaching in task-based settings. Wearables, which are capable of sound and visual interactions, are used by each student. An interface for input actions in the system when asked has been developed. The wearables are also used to track and log the students' interactions with objects (using NFC technology) and the students' location in the task-based setting (using Bluetooth technology). A visualization device (i.e. monitor) is also used to display the progress of the learning process in an engaging way. The results showed that this collaboration-supporting and IoT-based learning process was more effective. Ali and Hassan [17] used the Google Home and Google AI technologies to develop a voice-enabled system, which was used not only to assist the teaching process but as an assessing tool as well. Orally expressed questions can be understood and be followed by appropriate audio answers. Furthermore, answers can be listened, understood and assessed appropriately. With the utilization of this system, the students were able to interact with the learning subject matter in a more engaging way.

Tziortzioti et al. [18], in a STEM context, used microcontrollers and sensors capable of measuring water quality metrics to familiarize students with water quality and conservation concepts in a constructive manner. Similarly, an environmental centric approach is described in [19] by Pruet et al. They used a single board computer with multiple sensors and a cam as well as a monitoring child-friendly web interface to allow students to observe, experiment and draw conclusions about the correlation between weather conditions and plant development.

Barron [20] developed a wearable prototype for assisting aquatic physical education. This wearable device contains a system-on-chip and sensors that can sense, store and communicate biometric data of swimmers during training. Thus, the trainers can monitor the students' swimming style and give appropriate advices for improvement. Schafer [23] used the inverted classroom technic in combination with appropriate reading material and IoT hardware. The goal was to enable students have more time to perform practical C++ programming exercises based on the available devices (Raspberry Pi and SenseHAT). In [25], Ferreira et al. proposed a problem-based learning methodology for teaching IoT specific courses, using a three-phase methodology. The three iterative phases for designing an IoT system are: the understanding of the business, the definition of requirements and the implementation. Implementation solutions and guidance are available in each step ensuring at the same time the concept of learning by doing. The evaluation showed that this methodology facilitated the required goals and the students approved its adoption in future courses.

Jacko et al. [24] presented a way to realize a remote IoT laboratory where the students work remotely in real hardware and the teachers can monitor and intervene on the students' hardware. They created and used it in a microprocessor technology course. During the COVID-19 pandemic period, the students were able to practice safely from home on a real microcontroller (STM32) which was connected with a central server through an interfacing device (esp8266). The teachers via the use of monitoring software were able to observe and manage the students' interactions with their hardware. The evaluation results showed significant performance improvements and an increase in satisfaction among most of the students.

IV. Combination of Augmented Reality (AR) and Internet of Things (IoT) technologies in education

The amount of the research regarding the combined use of IoT and AR technologies in education is significantly smaller, a fact that signifies an open field for exploring innovation.

A smartphone application was created in [15] that can help users find their way around the university campus using AR hints, signs and direction arrows in cooperation with the developed university spatial guidance system. In [27], a way to perform safely gamma-radiation spectrum measurements in school laboratories was presented. A mobile application for presenting AR objects that the user can interact with, which are connected with real lab objects that can perform the virtually guided actions in the real lab, was used. An AR mobile application that assists students with the usage of informative visualizations during the use of Arduino technology is presented in [28].

Mylonas et al. [29] presented a system that utilizes AR in class activities pertinent to energy-efficiency in school buildings. AR was used for enhancing the way that students and teachers interact with the existent IoT system. The addition of AR enriches the visualization of the information offered, offers visual aids to help students complete the required labs and makes easier the management of the IoT

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system. In [26], AR and IoT teaching aids were created and evaluated. AR cards and an IoT device were used to monitor air pollution. In both approaches, besides the learning satisfaction positive results in learning performance were recorded as well.

V. AR and IoT Proposed Series of Activities

Students of the first class of senior high school have to be introduced and get familiar with a number of concepts that are included in the curriculum of the "Informatics Applications" course. Programming, mobile apps creation, web page creation, and cloud services are some of the concepts that students should become acquainted with. After examining related cases in the sections above, we have come up with the idea of creating a series of project-based activities to introduce students in IoT and AR technologies. These activities intend to lead students though the process of creating an IoT system that can be monitored and managed by AR technology. A series of processed steps ready to be applied will be guiding the students, while the curriculum guidelines will be followed as well. Technologies and new concepts will be explained at each step so as students would have a clear understanding of what they were doing and why they were doing it. Thus, the students at the end of the activities series will have a full overview of the technologies involved as well as a hands-on experience on implementing and creating such a system, which constitutes the innovation of this particular approach. In Figure 2, we present an outline of the proposed activities.

Fig. 2. IoT and AR proposed activities series.

Firstly, an overview of the scenario (e.g. monitoring and managing the school's garden) will be presented to the students. Questions like "how the system will be functioning?" as well as "what technologies are involved?" will be answered. The next activity will be about the IoT system itself. The students should become familiar with concepts such as microcontrollers, sensors, actuators and should be introduced to the selected for the activities microcontroller. The diagram of the system that the students will have to realize, as well as the required electronic parts, will be presented to and discussed with students.

The students will proceed and complete the assembly of the parts of the IoT system following the steps that will have been given to them in advance. The programming of the microcontroller will also be included in this activity. It will contain the uploading of the code, a brief explanation of the code, a demonstration of the working code and debugging of the students' code. Also, the creation of a web page or a mobile application for monitoring the sensed metrics can be some scenarios of what the uploaded code might be capable of achieving. The testing of the system, according to the appropriate guidelines, will ensure the successful completion of the IoT system.

Having the IoT system completed, the students will be briefly introduced to a 2D-3D design and an AR development environment. The students will open an existing project and follow a series of actions in order to create and setup the dashboard for monitoring and managing the IoT system. The testing and the debugging of the AR project will also ensure the overall functionality of the AR IoT system, concluding the activities.

Following the above-mentioned activities, the students will have a chance to get familiar with informatics concepts using up-to-date technologies and tools. Thus, a more interesting context is offered, facilitating a more efficient learning and teaching process.

VI. Conclusions and Further Research

It is a fact that Augmented Reality (AR) technologies can engage students more actively in the learning process and they can assist in the strengthening of the interactions between them. By exploiting the AR technology possibilities the students can be inspired to engage in the learning process in an increased level through the experience of immersion [45][46].

In addition, IoT systems in education offer a series of advantages. Remote and safer labs, more and better assessment criteria through the detailed monitoring and logging of students' actions, more efficient learning context via capability of observing real world metrics, a field to experiment and apply knowledge for ICT courses are some of the advantages in IoT use cases in the educational practice.

A series of activities for creating an IoT and AR-based system was discussed in this paper so as to create a proposal of an alternative way of dealing with the requirements of the curriculum of the "Informatics Applications" course. Thus, a more concrete way of addressing concepts that are usually introduced in a more abstract and out of date way, is proposed through this work. So far, we work on creating and organizing the required material (e.g. worksheets, electronic components diagram, code and virtual objects) for the proposed steps of activities. After that, we are going to implement it into the educational process and receive the teachers and students' opinions.

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References

[1] Arghashi, V., Yuksel, C. A. (2022). Interactivity, Inspiration, and Perceived Usefulness! How retailers' AR-apps improve consumer engagement through flow. Journal of Retailing and Consumer Services, 64, 102756.

[2] Poh, J. J., Phua, S. X., Chan, K. F., Gan, S. K. E. (2018). Commentary: Augmented Reality Scientific Phone Apps–making the APD AR Holistic Review app and using existing AR apps for scientific publications. Sci Phone Apps Mobile Devices, 4(4).

[3] Do, H. N., Shih, W., Ha, Q. A. (2020). Effects of mobile augmented reality apps on impulse buying behavior: An investigation in the tourism field. Heliyon, 6(8), e04667.

[4] Almoosa, A. (2015, October). The era of BYOD: Augmented reality apps in higher education. In E-Learn: World Conference on E-Learning in Corporate, Government, Healthcare, and Higher Education (pp. 1684-1689). Association for the Advancement of Computing in Education (AACE).

[5] Nikhashemi, S. R., Knight, H. H., Nusair, K., Liat, C. B. (2021). Augmented reality in smart retailing: A (n)(A) Symmetric Approach

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to continuous intention to use retail brands' mobile AR apps. Journal of Retailing and Consumer Services, 60, 102464.

[6] Semtech Corp. (n.d.). Lora technology is connecting our smart planet. Semtech. Retrieved May 18, 2022, from https://www.semtech.com/lora/lora-applications

[7] Leong, Y. M., Letchumanan, C. (2019, September). Effective learning in higher education in Malaysia by implementing internet of things related tools in teaching and introducing IoT courses in curriculum. In 2019 1st IEEE International Conference on Artificial Intelligence and Data Sciences (AiDAS), 152-157.

[8] Diddeniya, S. A. I. P., Gunawardana, H. D. C. N., Maduwantha, K., Koswattage, K. R., Randima, M. V., Vasanthapriyan, S. (2020). IoT based energy efficient smart classroom. J. Multidiscip. Eng. Sci. Stud, 6(12), 3581-3586.

[9] Mataloto, B., Ferreira, J. C., Cruz, N. (2019). LoBEMS—IoT for building and energy management systems. Electronics, 8(7), 763.

[10] Ahmed, A., Parvez, M. R., Hasan, M. H., Nur, F. N., Moon, N. N., Karim, A., ... Jonkman, M. (2019, January). An intelligent and secured tracking system for monitoring school bus. In 2019 IEEE International Conference on Computer Communication and Informatics (ICCCI), 1-5.

[11] Kumari, M., Kumar, A., Khan, A. (2020, February). IoT based intelligent real-time system for bus tracking and monitoring. In 2020 IEEE International Conference on Power Electronics & IoT Applications in Renewable Energy and its Control (PARC), 226-230.

[12] Elbasi, E. (2020, December). Reliable abnormal event detection from IoT surveillance systems. In 2020 IEEE 7th International Conference on Internet of Things: Systems, Management and Security (IOTSMS), 1-5.

[13] Yang, S., Song, Y., Ren, H., Huang, X. (2016, August). An automated student attendance tracking system based on voiceprint and location. In 2016 IEEE 11th International Conference on Computer Science & Education (ICCSE), 214-219.

[14] Mahmood, S., Palaniappan, S., Hasan, R., Sarker, K. U., Abass, A., Rajegowda, P. M. (2019, January). Raspberry PI and role of IoT in Education. In 2019 4th IEEE MEC International conference on big data and smart city (ICBDSC), 1-6.

[15] Torres-Sospedra, J., Avariento, J., Rambla, D., Montoliu, R., Casteleyn, S., Benedito-Bordonau, M., ... & Huerta, J. (2015). Enhancing integrated indoor/outdoor mobility in a smart campus. International Journal of Geographical Information Science, 29(11), 1955-1968.

[16] De La Guia, E., Camacho, V. L., Orozco-Barbosa, L., Brea Lujan, V. M., Penichet, V. M. R., Lozano Perez, M. (2016). Introducing IoT and Wearable Technologies into Task-Based Language Learning for Young Children. IEEE Transactions on Learning Technologies, 9(4), 366–378.

[17] Ali, M., & Hassan, A. M. (2019). Developing Applications for Voice Enabled IoT Devices to Improve Classroom Activities. 2018 21st International Conference of Computer and Information Technology, ICCIT 2018, 21–23.

[18] Tziortzioti, C., Mavrommati, I., Kalkavouras, C., Dimitriou, E., Chatzigiannakis, I. (2019, September). Observation and analysis of environmental factors of surface waters: An internet of things educational approach. In 2019 IEEE First International Conference on Societal Automation (SA), 1-7.

[19] Pruet, P., Ang, C. S., Farzin, D., Chaiwut, N. (2015, June). Exploring the Internet of "Educational Things"(IoET) in rural underprivileged areas. In 2015 12th IEEE International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology (ECTI-CON), 1-5.

[20] Han, B. (2019). Design and Implementation of Wearable Devices to Enhance Aquatics Physical Education. International Association of Societies of Desing Research Conference, 1-18.

[21] Krishnamurthi, R. (2018, August). Teaching methodology for iot workshop course using node-red. In 2018 Eleventh IEEE International Conference on Contemporary Computing (IC3), 1-3.

[22] Silvis-Cividjian, N. (2019, May). Teaching Internet of Things (IoT) literacy: A systems engineering approach. In 2019 IEEE/ACM 41st IEEE International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET). 50-61.

[23] Schäfer, U. (2019, April). Teaching modern C++ with flipped classroom and enjoyable iot hardware. In 2019 IEEE Global Engineering Education Conference (EDUCON), 910-919.

[24] Jacko, P., Bereš, M., Kováčová, I., Molnár, J., Vince, T., Dziak, J., ... Kováč, D. (2022). Remote IoT Education Laboratory for Microcontrollers Based on the STM32 Chips. Sensors, 22(4), 1440.

[25] Ferreira, L. C. B. C., Branquinho, O. C., Chaves, P. R., Cardieri, P., Fruett, F., Yacoub, M. D. (2019). A PBL-based methodology for IoT teaching. IEEE Communications Magazine, 57(11), 20-26.

[26] Liao, Y. W., Hsieh, M. C., Wei, C. W. (2021, July). Effectiveness of Integrating AR and IoT technologies into Environmental Education for Elementary School Students. In 2021 IEEE International Conference on Advanced Learning Technologies (ICALT), 78-80.

[27] Caputo, E., Moriello, R. S. L., de Pandi, F., Liccardo, A., Gallicchio, V., Quarto, M. (2021, May). Exploiting augmented reality and internet of things for gamma ray experiments in educational field. In 2021 IEEE International Instrumentation and Measurement Technology Conference (I2MTC), 1-6.

[28] Panessai, I. Y., Iksan, N., Zahari, S. A., Abdulbaqi, A. S., Modi Lakulu, M. M. B., Husin, M. R., ... Wahyudiono, P. H. (2021, March). Learning Internet of Things by using Augmented Reality. In 2021 the 5th International Conference on Virtual and Augmented Reality Simulations, 17-20.

[29] Mylonas, G., Triantafyllis, C., Amaxilatis, D. (2019). An augmented reality prototype for supporting IoT-based educational activities for energy-efficient school buildings. Electronic Notes in Theoretical Computer Science, 343, 89-101.

[30] Serin, H. (2017). Augmented Technologies in the education: AR apps. Journal of Educational Sciences & Psychology, 7(2).

[31] Martín-Gutiérrez, J., Mora, C. E., Añorbe-Díaz, B., González-Marrero, A. (2017). Virtual technologies trends in education. Eurasia Journal of Mathematics, Science and Technology Education, 13(2), 469-486.

[32] Clarke, N. I. (2019). Through the screen and into the world: augmented reality components with MIT App Inventor (Doctoral dissertation, Massachusetts Institute of Technology).

[33] Lytridis, C., Tsinakos, A. (2018). Evaluation of the ARTutor augmented reality educational platform in tertiary education. Smart Learning Environments, 5(1), 1-15.

[34] O'Shea, P. M., Elliott, J. B. (2016, June). Augmented reality in education: An exploration and analysis of currently available educational apps. In International Conference on Immersive Learning (pp. 147-159). Springer, Cham.

[35] Hordiienko, V. V., Marchuk, G. V., Vakaliuk, T. A., Pikilnyak, A. V. (2020). Development of a model of the solar system in AR and 3D. Proceedings of the 3rd International Workshop on Augmented Reality in Education, 217-238.

[36] Alfaro, J. L. D., Puyvelde, P. V. (2021, September). Mobile Augmented Reality Apps in Education: Exploring the User Experience Through Large-Scale Public Reviews. In International Conference on Augmented Reality, Virtual Reality and Computer Graphics (pp. 428-450). Springer, Cham.

[37] Tanalol, S. H., Baharum, A., Ahmed, M. M., Tahir, A., Saudi, A., Wahab, S. H. A., Saad, N., Foo, J. Development of Augmented Reality Mobile Apps for STEM Education. Transactions on Science and Technology, 8(3-3), 589 – 598.

[38] Suhaimi, S. B., Yatim, N. A. B. M., Abd Wahab, M. H., Idrus, S. Z. S. (2020, April). Jawi-AR Apps: Single Jawi Character Connecting Method to single words using AR Technology. In Journal of Physics: Conference Series (Vol. 1529, No. 3, p. 032001). IOP Publishing.

[39] Fidan, M., Tuncel, M. (2019). Integrating augmented reality into problem based learning: The effects on learning achievement and attitude in physics education. Computers & Education, 142, 103635.

[40] Auliya, R. N., Munasiah, M. (2020). Augmented Reality Affects Students' Attitude and Conceptual Understanding in Learning 3D Geometry. JPI (Jurnal Pendidikan Indonesia), 9(2), 203-212.

[41] Luna, U., Rivero, P., Vicent, N. (2019). Augmented reality in heritage apps: Current trends in Europe. Applied Sciences, 9(13), 2756.

[42] Lytridis, C., Tsinakos, A., Kazanidis, I. (2018). ARTutor—an augmented reality platform for interactive distance learning. Education Sciences, 8(1), 6.

[43] Oracle Corp. (n.d.). What is IoT? Retrieved May 20, 2022, from https://www.oracle.com/internet-of-things/what-is-iot/

[44] Internet Engineering Task Force (IETF) (n.d.). The Internet of Things at the IETF. Retrieved May 20, 2022, from https://www.ietf.org/topics/iot/

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[45] Ahsan, M. G. K., & Cahyono, A. N. (2020, June). Designing augmented reality-based mathematics mobile apps for outdoor mathematics learning. In Journal of Physics: Conference Series (Vol. 1567, No. 3, p. 032004). IOP Publishing.

[46] Lindner, C., Rienow, A., Jürgens, C. (2019). Augmented Reality applications as digital experiments for education–An example in the Earth-Moon System. Acta Astronautica, 161, 66-74.

[47] ARToolKit, available at: http://www.artoolkitx. org/.

[48] ARCore, available at: https://developers.google.com/ar

[49] Unity, available at: https://unity.com/

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